TASK 3: SQL DATA ANALYSIS

1.Introduction

SQL was used to perform business analysis and extract valuable insights from the cleaned dataset.

2.Objective

- Calculate key business KPIs.
- Analyse revenue performance.
- Identify best-performing products.
- Analyse referral source contribution.

3.Tools Used

- SQLite
- SQL Queries

4.SQL Analysis Performed

Query 1: Total Orders

```
SQL Query:

SELECT COUNT(*)-1 AS total_orders
FROM orders;

Result: 1200
```

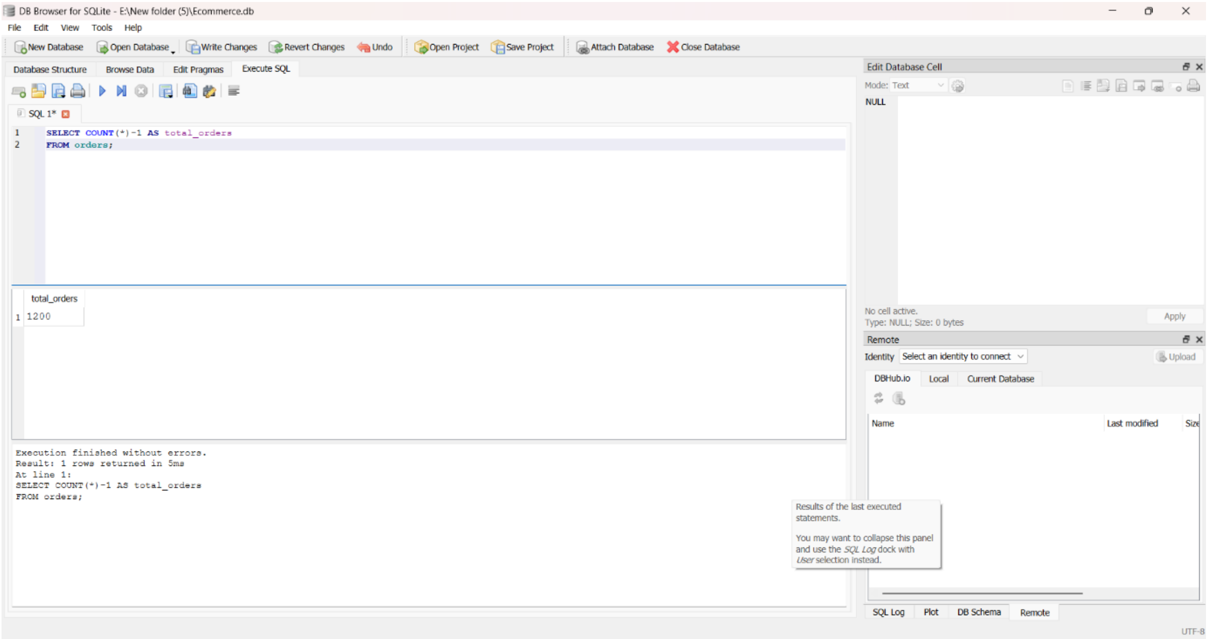

Figure 1: Total Orders Query Output

Query 2: Total Revenue

SQL Query:

```
SELECT ROUND(SUM(CAST(field15 AS REAL)),2) AS total_revenue
FROM orders
WHERE field15 != 'TotalPrice';
```

Result: ₹12,64,761.96

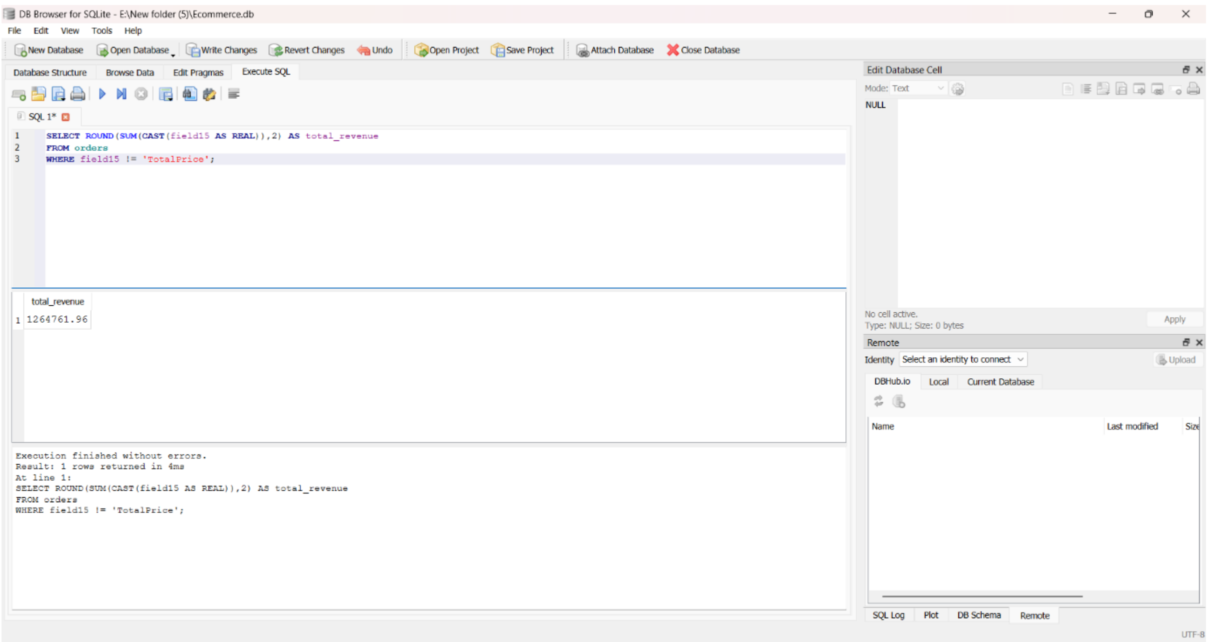

Figure 2: Total Revenue Query Output

Query 3: Average Order Value

SQL Query:

```
SELECT ROUND(AVG(CAST(field15 AS REAL)),2) AS avg_order_value
FROM orders
WHERE field15 != 'TotalPrice';
```

Result: ₹1053.97

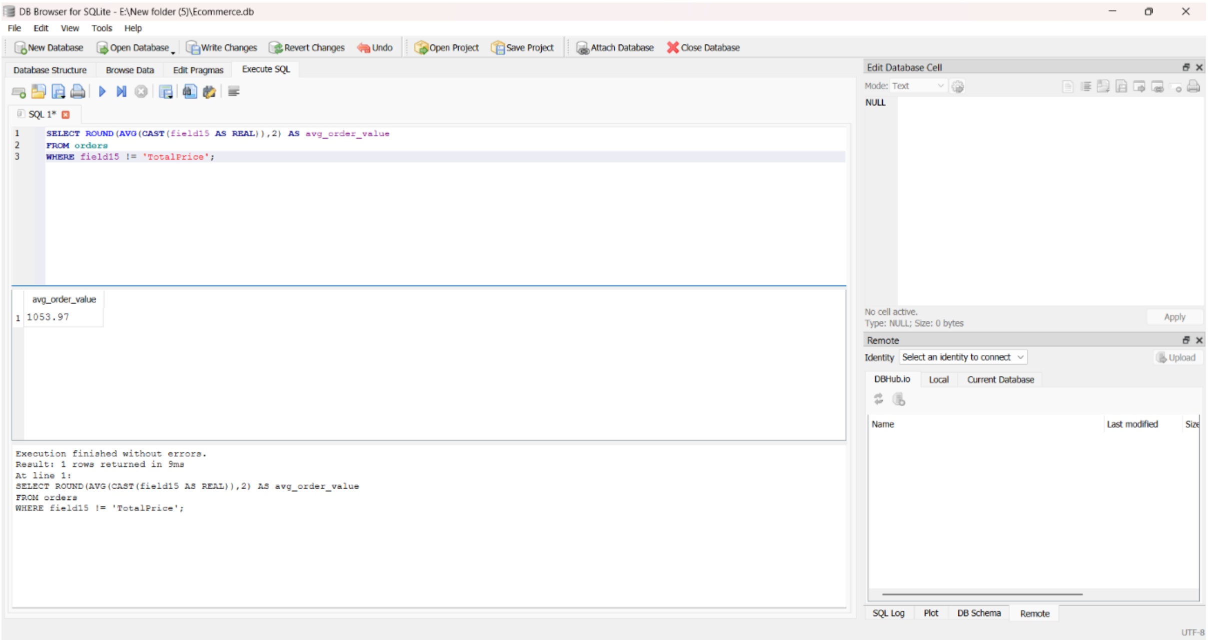

Figure 3: Average Order Value Query Output

Query 4: Product Revenue Analysis

SQL Query:

```
SELECT field5 AS Product,
ROUND(SUM(CAST(field15 AS REAL)),2) AS Revenue
FROM orders
WHERE field15 != 'TotalPrice'
GROUP BY field5
ORDER BY Revenue DESC;
```

Result:

Chair → ₹195,620.11
Printer → ₹195,612.61
Laptop → ₹192,126.56

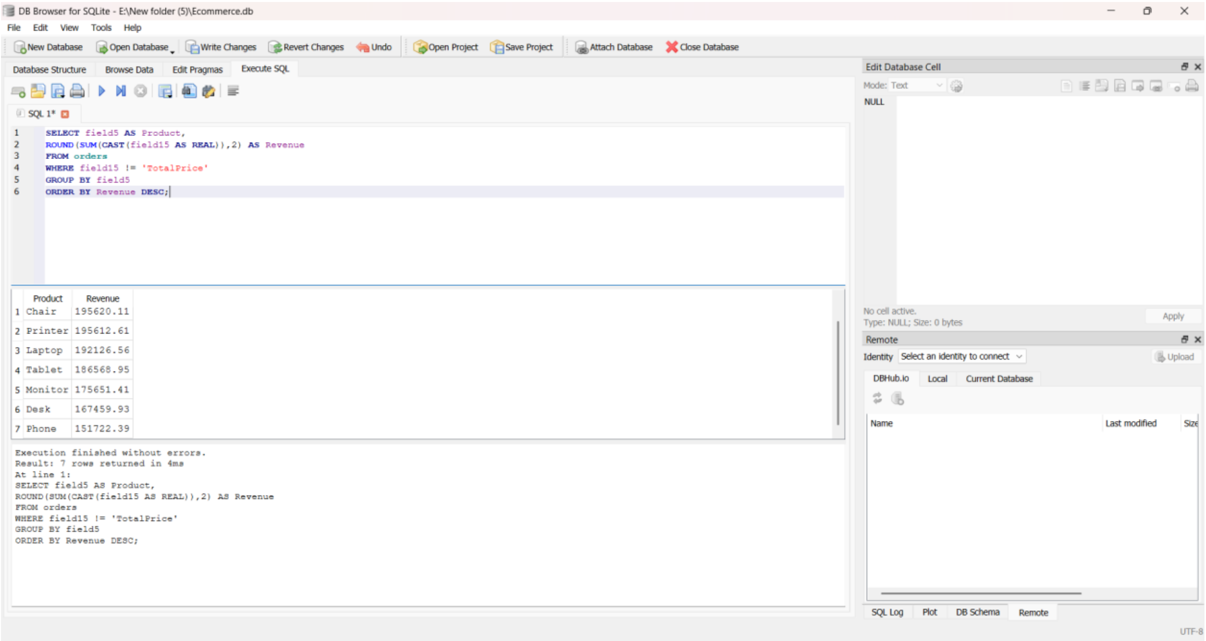

Figure 4: Product Revenue Query Output

Query 5: Best Product

SQL Query:

```
SELECT field5 AS Product,
ROUND(SUM(CAST(field15 AS REAL)),2) AS Revenue
FROM orders
WHERE field15 != 'TotalPrice'
GROUP BY field5
ORDER BY Revenue DESC
LIMIT 1;
```

Result:

Best Product = Chair
Revenue = ₹195,620.11

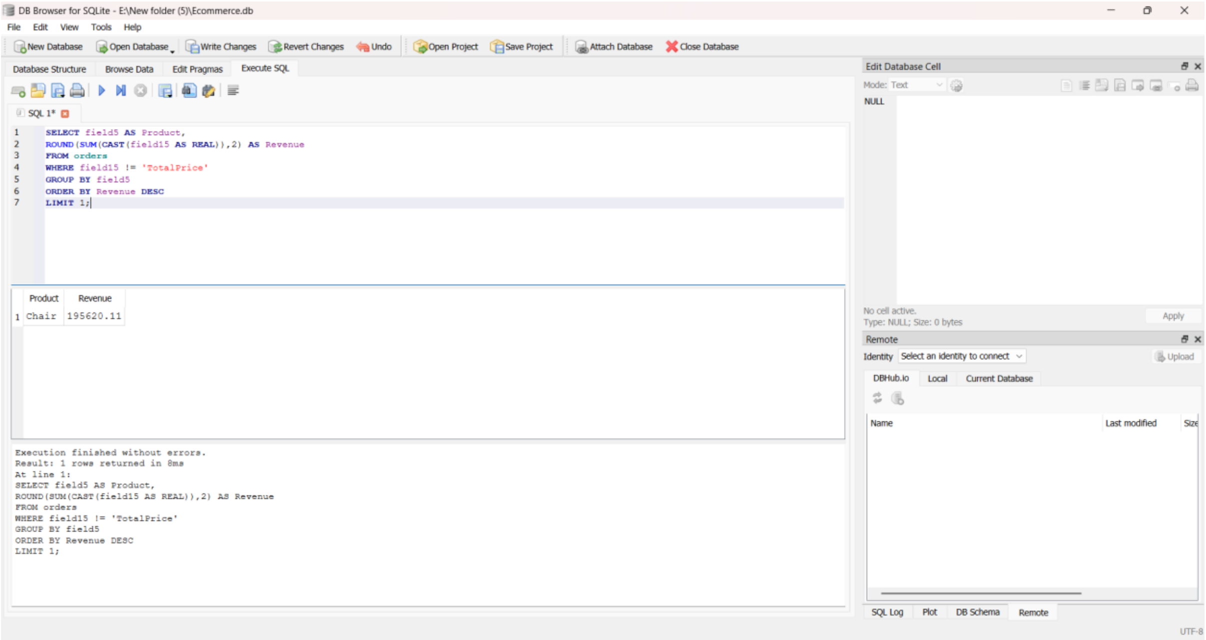

Figure 5: Best Product Query Output

Query 6: Referral Source Analysis

SQL Query:

```
SELECT field14 AS ReferralSource,
ROUND(SUM(CAST(field15 AS REAL)),2) AS Revenue
FROM orders
WHERE field15 != 'TotalPrice'
GROUP BY field14
ORDER BY Revenue DESC;
```

Result:

Instagram → ₹275,285.45
Email → ₹261,808.55
Google → ₹250,441.48

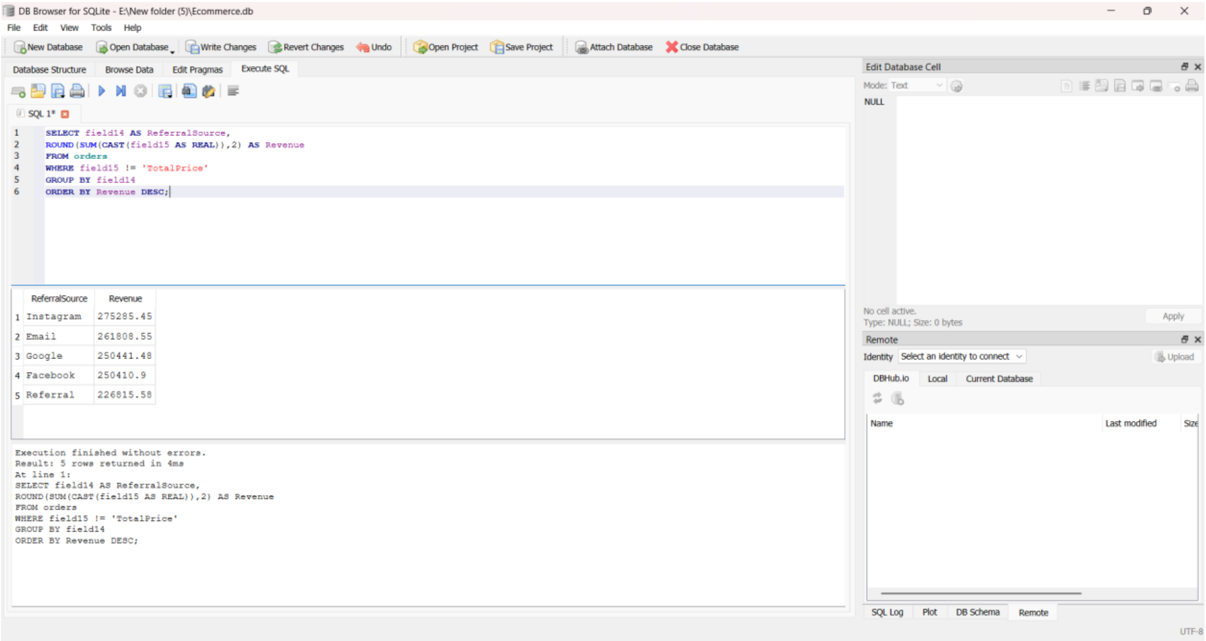

Figure 6: Referral Source Analysis Output

Query 7: Best Referral Source

SQL Query:

```
SELECT field14 AS ReferralSource,
ROUND(SUM(CAST(field15 AS REAL)),2) AS Revenue
FROM orders
WHERE field15 != 'TotalPrice'
GROUP BY field14
ORDER BY Revenue DESC
LIMIT 1;
```

Result:

Instagram → ₹275,285.45

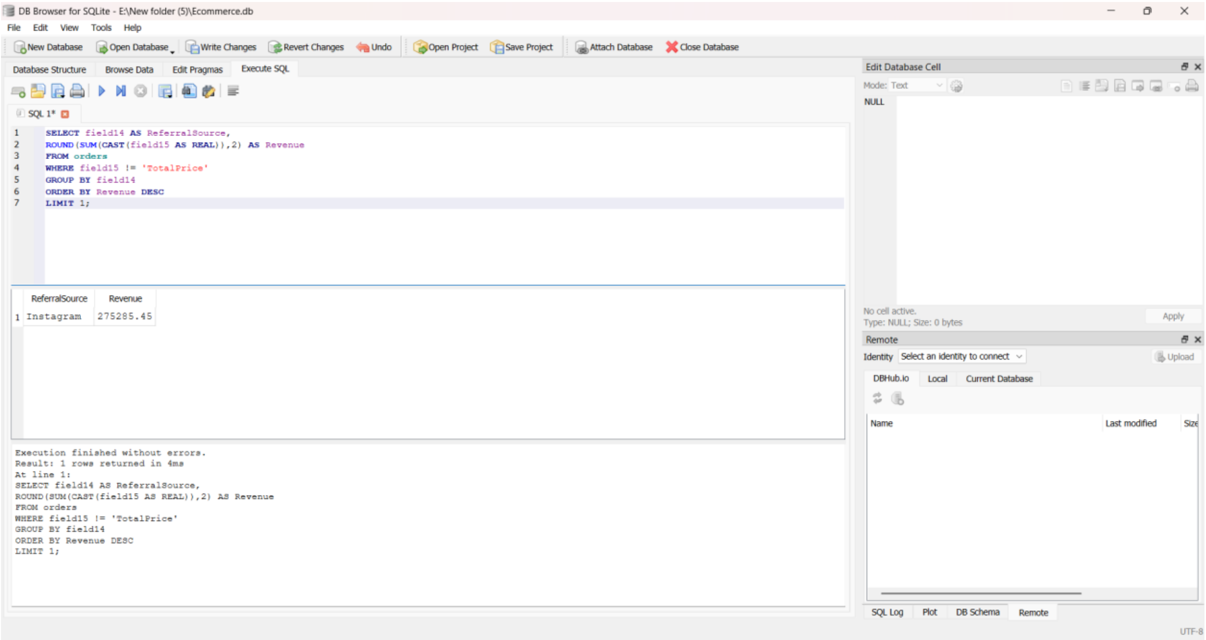

Figure 7: Best Referral Source Query Output

Query 8: Revenue by Year

SQL Query:

```
SELECT SUBSTR(field2,1,4) AS Year,
ROUND(SUM(CAST(field15 AS REAL)),2) AS Revenue
FROM orders
WHERE field15 != 'TotalPrice'
GROUP BY Year
ORDER BY Revenue DESC;
```

Result:

2023 → ₹552,643.24
2024 → ₹480,235.87
2025 → ₹231,882.85

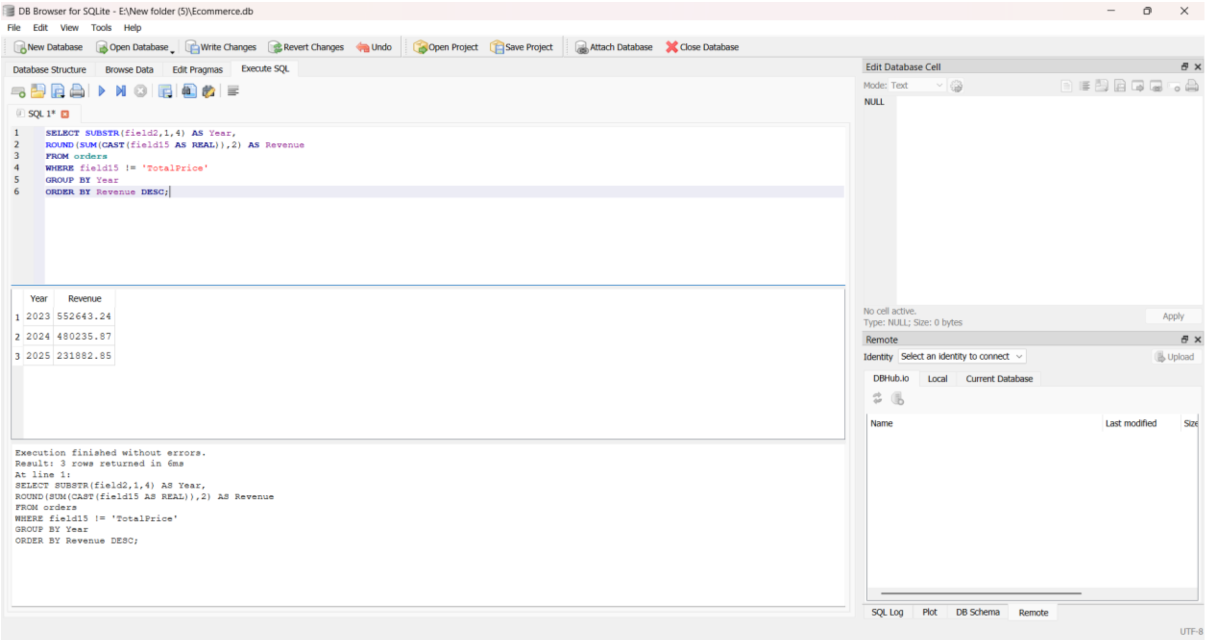

Figure 8: Revenue by Year Query Output

Query 9: Best Month

SQL Query:

```
SELECT SUBSTR(field2,1,7) AS Month,
ROUND(SUM(CAST(field15 AS REAL)),2) AS Revenue
FROM orders
WHERE field15 != 'TotalPrice'
GROUP BY Month
ORDER BY Revenue DESC
LIMIT 1;
```

Result:

June 2024 → ₹68,068.54

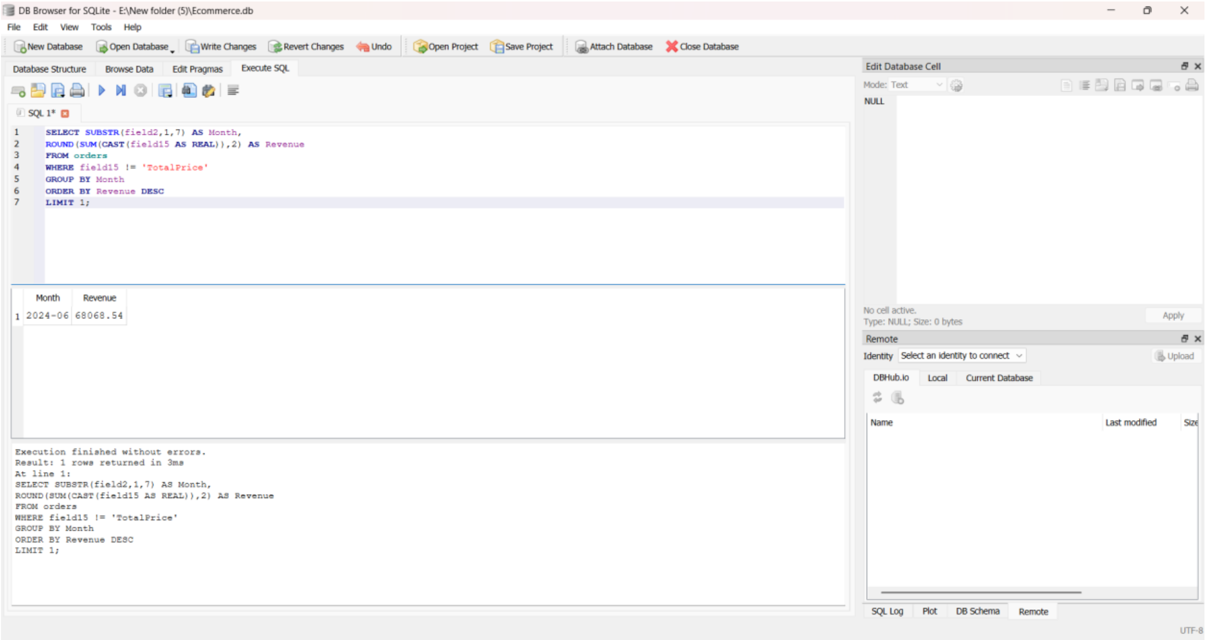

Figure 9: Best Month Query Output

5.Key Insights

- Chair is the highest revenue-generating product.
- Instagram is the strongest customer acquisition channel.
- Revenue exceeds ₹12.64 Lakhs.

6.Conclusion

SQL helped transform raw transactional data into actionable business insights.